PROJECT REPORT

Project Name	NexusML	Reporting Period
Project Owner	Viren Singh	Sep 18, 2026 to Sep 22, 2026
Prepared by	Viren Singh	

HIGHLIGHTS

- Built this **AI risk engine that looks at multiple factors** to predict urban disasters.
- **Tried three models inside it** – logistic regression, a weighted one, and then an ensemble. Honestly the ensemble was kind of a pain to get working but it ended up being the most useful
- **Integrated uncertainty quantification** using the Monte Carlo simulation to estimate prediction stability
- Built **real time drift monitoring system** to detect data distribution shifts and model degradation
- Added **Explainable AI (XAI) module to identify top contributing factor** for each prediction in the programm
- **Enabled counterfactual scenario** simulation to analyze how changes... in environment and infrastructure can affect the value of risk
- Developed a fully interactive dashboard **based ML-system** with real time scenario evaluation

CHALLENGES

- Designed a unified risk scoring system that bring together multiple models along with the uncertainty, anomaly detection , and drift signals into a single consistent framework.
- Worked on balancing interpretability with complexity so the system stays explainable while still capturing advance ML concepts.
- Simulated a realistic data drift without relying on the real time external dataset or live data streams
- Ensured consistency across different models output while comparing the logistic , weighted ,and ensemble - based approaches.
- Made sure that counterfactual simulation produce realistic and logically consistent change the risk value so that the program can update fast on the user input.
- Structured everything into a single interactive dashboard while keeping it responsive and easy to understand.

Live project : link

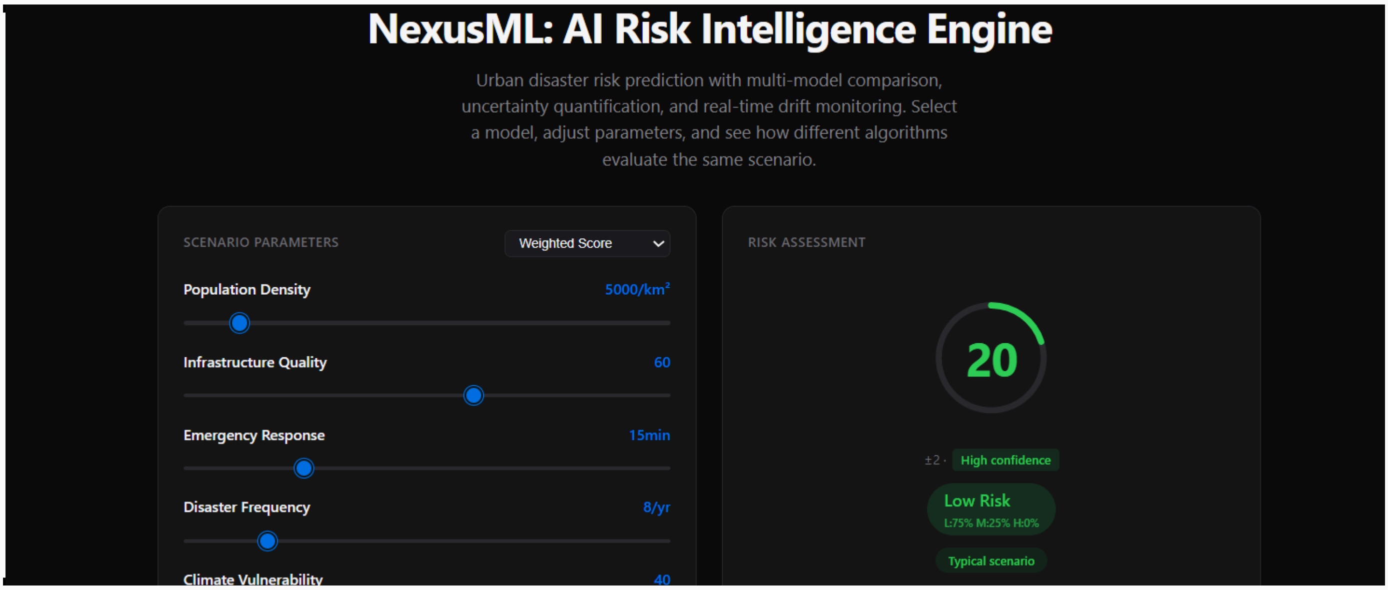
https://virensingh011.github.io/NexusML/

Github Account :

https://github.com/virensingh011

System showcase:- screenshot

Dashboard



- **Interactive interface to keep things easy to understand** and to adjust urban disaster risk parameters in real time

- It **uses multi-factor inputs** including population density, climate, infrastructure quality, and more.

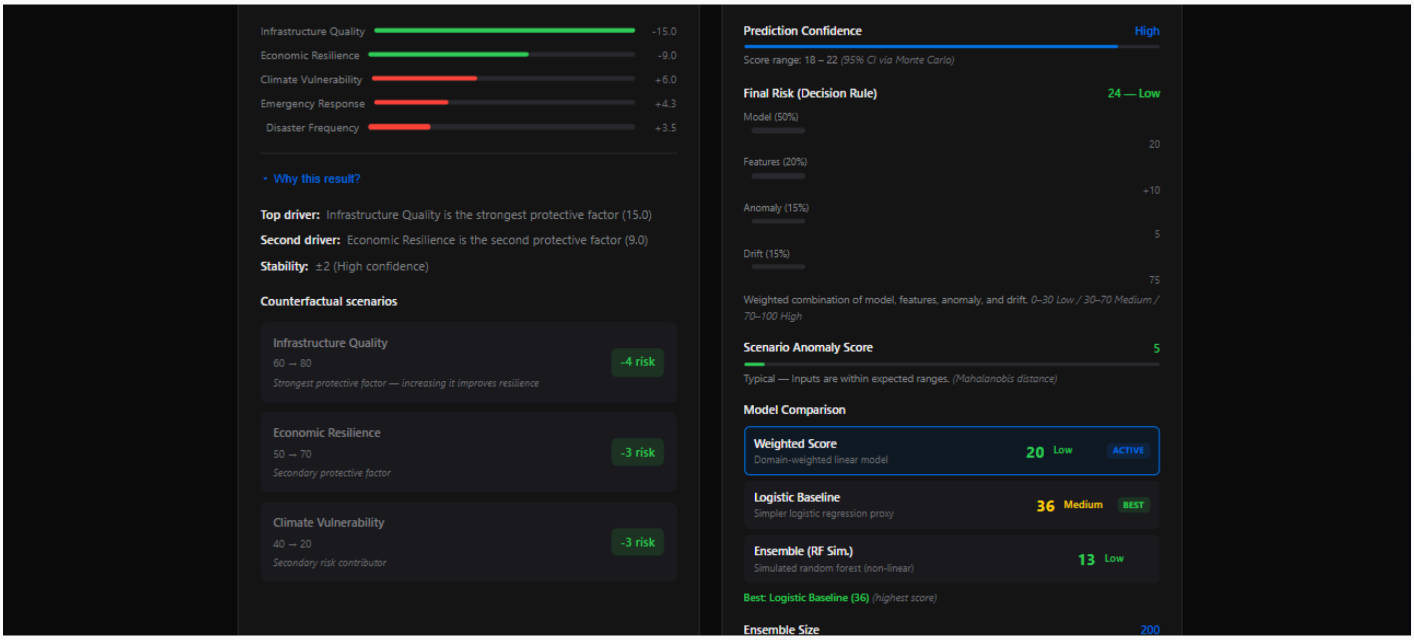
- **Computes risk score** using a weighted evaluation model

- **Provides probability distribution** of risk outcomes (L / M / H%).

- **Shows real-time updates of risk based** on parameter changes.

- **Enables scenario-based analysis** for understanding impacts of environmental changes.

• AI Risk Evaluation Engine



- **Shows how each input factor contributes** to final risk prediction in a easy to understand way.

- It **highlights the most important factor** contribute to each decision which help in understanding what are the factor if enhance so to reduce the risk of problem & disaster

- **Combines model output** , features impact and drift signals into one final risk result.

- **It compare different models** (weighted , logistic , Ensemble) to check the consistency in result.

- **Detect unusual input patterns** using anomaly scoring.

- **It monitors data drift** to identify when the system start behaving differently from expected pattern this help the system to fix itself and advise user to reload.

- **Uses Monte Carlo simulation** to test stability under small variation in inputs.
- **Updates risk results instantly** when parameters are changed so that help in understanding fast

Scenario Comparison

- Scenario Comparison is a feature **also user to try different possibility** and analysis how does the different parameter affect the value

- It **enable adjustment of key urban parameters** such as population ,density and more

- It **simulates how changes in inputs affect overall risk level** in the condition

- It **include in two scenario** – scenario 1 and 2 this help to study the two different parameter at single time in a single screen

- **Support the data drift simulation** to show how model behavior changes under different inputs distributions.

- **Provide a clear comparison between the two scenario** A and scenario B for better decision making

GUIDE : HOW TO USE THE NEXUSML

- **Step 1**: Open the project by clicking the link given in the report, which will launch the NexusML dashboard in your browser.

- **Step 2**: Wait for the dashboard to load properly and get familiar with the main interface.

- **Step 3**: Go to the Scenario Parameters section and adjust values like population density, infrastructure quality, climate vulnerability, and other factors.

- **Step 4**: Watch the risk score update in real time as you change the inputs.

- **Step 5**: Try switching between different models (Logistic, Weighted, Ensemble) to see how each one evaluates the same scenario differently.

- **Step 6**: Scroll down to the Feature Influence section to understand which factors are affecting the result the most.

- **Step 7**: Check the Prediction Confidence and Uncertainty section to see how stable the model's output is.

- **Step 8**: Use the Counterfactual Simulation tool to change values and observe how the risk changes in different conditions

- **Step 9**: Open the Data Drift section to see whether the current inputs match expected patterns or show any deviation.

- **Step 10**: Use the Scenario Comparison tool to compare different cases side by side and analyze differences in risk.

- **Step 11**: Finally, review the Risk Dashboard, where all signals come together into a single final decision output.

nexusML

Thank You

This project, (NexusML) is an AI Risk Intelligence Engine, demonstrates an approach to combine the multiple machine learning techniques for urban disaster risk analysis. Which highlights the importance of interpretability, uncertainty estimation, and realtime monitoring in decision-making systems.

Thank you for reviewing this project.